

Operativity and Representativity of the Sign in Leibniz

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Abstract In opposition to, or in the line of, philosophy cannot grow except inside a tradition. This is precisely the issue with contemporary research on diagrammatic thinking. We all agree that it is essential to go back to Peirce or Husserl not only to recognize important roots, to honour significant predecessors, but to recover their insights, to make use of their hypothesis, to recuperate the conceptual instruments that they have been able to put forward. What I propose here is to go back even more in time, to go back to Leibniz. Not to claim for Leibniz as a predecessor of cognitive semantics or of present-day diagrammatic research, but to recall what Leibniz thought out in this respect, to bring into play the conceptual devices he put up, to bear in mind the distinctions he was able to constitute.

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This paper is a homage to Jean-Yves Béziau and to his marvellous capacity of combining the strictness and rigor of logics with the care and compassion of meaning.

1 Leibniz' Cognitive Conception of Language

My first claim is that the historical importance of Leibniz' philosophy of language is mostly due to his cognitive conception of language, that is, his recognition of the constitutive symbolic nature of human thought. As he says in the celebrated *Dialogus de connexione inter res et verba* (1677):

I will never be able to know, to discover, to prove without using words or without the presence in my mind of other signs. [2, 7.191]

This cognitive conception of language was mainly formulated in the scope of the debate with Descartes and of their different mathematical experiences. Descartes tends to star in geometry, in which figural representation is a merely auxiliary, imaginative support for reasoning. Leibniz stars in algebra where operations are fully symbolically performed.

where symbols constitute the reasoning process itself, completely replacing the supposed direct experience of mathematical objects.

In fact, the Cartesian principle of evidence entails a merely instrumental and communicative conception of language. For Descartes, language has above all a communicative function. The most language may be asked to do is to operate as a mnemonic support for the recall of the long chain of reasons. Leibniz refuses the intuitionism of Descartes. For Leibniz, rigour cannot be dependent either on subjectively based certainties or on the confidence in the intuitive infallibility of natural light. As Leibniz says in a *Letter to Gallois* (1677),

[The Cartesian methodological rules] give surely beautiful decrees but not the guideline for developing those decrees. [2, 7.21]

According to Leibniz, rigour should be achieved by the use of a symbolic system which would render visible the more abstract ideas and would constitute a material support for thought and reasoning, a "*filum Ariadnes*" [2, 7.22]¹, a "*filum palpabile*" [2, 7.57, 59, 125], a *filum cogitandi* [1, 420], a *filum meditandi* [2, 7: 14], a *filum mechanico* [1, 351], that is, a symbolic criterion, a manipulatory device. Something which mathematics has already developed and which, as Leibniz used to say, needs nothing but paper and ink. Let us look for what Leibniz writes in the well-known *Preface à la Science Générale* (1677)

Now the reason why the art of demonstrating has been until now found only in mathematics [...] is this: mathematics carries its own test with it. For when I am presented with a false theorem, I do not need to examine or even to know the demonstration, since I shall discover its falsity a posteriori by means of an easy experiment that is, by a calculation, costing no more than paper and ink.

The tests or experiments made in mathematics to guard against mistakes in reasoning [...] are not made on a thing itself, but on the characters which we have substituted in place of the thing.

Take for example a numerical calculation: if 1677 times 365 are 612,105, we should hardly ever have reached this result if it were necessary to make 365 piles of 1677 pebbles each and then finally to count them all in order to know whether the aforementioned number is found. This test is performed only on paper, and consequently on the characters which represent the thing, and not on the thing itself. [1, 154]

Symbolism is for Leibniz the proper, the necessary and essential means of human reason. Leibniz even goes a step further by claiming that only symbolic signs allow us to operate with ideal significations, which can only be established by the sign and that only signs enable us to think. The example of the polygon of a thousand faces presented by Leibniz in his celebrated *Méditationes de Cognitione, Veritate et Ideis* (1684) is eloquent:

When I think of a polygon of thousand faces, I do not always consider what is a face, an equality or the number thousand but I use this words (whose meaning is present on my spirit only in a very confuse and imperfect way) in order that they (the words) take the place of the ideas which I have of them [...] I call this knowledge as blind or symbolic. We make use of it in algebra and arithmetic and in almost all domains. [2, 4.423]

Again, Leibniz' mathematical experience lies in the basis of this important theory of blind thought (*cogitatio caeca*). In infinitesimal calculus, sign does not evoke but fully substitutes notions which the human mind cannot otherwise completely reach. So, I believe, we

¹ "La véritable méthode nous doit fournir un *Filum Ariadnes*, c'est à dire, un certain moyen sensible et grossier, qui conduise l'esprit, comme sont les lignes tracées en géométrie et les formes des opérations qu'on prescrit aux apprentis en Arithmétique" [2, 7.22].